How Many People Live Near Protected Areas in Developing Countries? Estimates from Gridded Population Data (2000–2020)

Abstract

Impact evaluations of protected areas routinely define "local populations" through spatial proximity, yet the demographic scale implied by these definitions is almost never quantified. We estimate population counts inside protected areas and within 10 km of their boundaries for 75 low- and lower-middle-income countries (LMICs) in 2000 and 2020, using the World Database on Protected Areas and globally harmonized gridded population data (GHSL, with WorldPop as robustness check). India is excluded because its national protected areas are not publicly available in the WDPA. Results are decomposed by IUCN management category (strict and non-strict). In 2020, approximately 61 million people (2.5% of their combined population) lived inside protected areas and over 900 million (36.6%) lived inside or within 10 km of their boundaries. These figures are substantially larger than those estimated for 2000, reflecting both PA expansion and demographic growth. Most PA-adjacent populations are located near non-strict protected areas (IUCN categories IV to VI) rather than strict protection regimes (IUCN categories I to III). These population orders of magnitude are relevant for interpreting the scope of PA impact evaluations but do not themselves constitute impact estimates.

~~Protected areas (PAs) are the mainstream biodiversity conservation tool. While the socioeconomic impacts of conservation policies remain debated, particularly in developing countries where poverty is predominantly rural, the number of people affected by PAs has never been systematically quantified. We estimate the population living within and near PAs from 2000 to 2020 in 76 countries classified as low- and lower-middle-income by the World Bank. Using the World Database on Protected Areas and two high-resolution gridded population datasets, we find that terrestrial PA coverage in these countries increased from 9.3% to 12.9%, while the share of the population living within 10 km of a PA rose from 16.4% to 21.8%. Excluding India, where PA expansion was minimal, coverage increased from 10.2% to 14.1%, and proximate populations from 26.5% to 33.5%. This substantial demographic expansion challenges the traditional view of conservation as isolated and affecting only small population groups. Policymakers should recognize conservation as a broad societal issue and integrate poverty reduction strategies into conservation planning and governance, particularly as global efforts accelerate towards protecting 30% of terrestrial land by 2030.~~

Introduction

Protected areas (PAs) were primarily~~are a primary tool for biodiversity conservation,~~ established ~~mainly~~ to achieve ecological purposes~~goals~~ (Maxwell et al. 2020). Their implications for human~~However, their creation can have positive or negative effects on local~~ populations ~~(Kandel et al. 2022).~~have nevertheless been a persistent concern in conservation research ~~PAs restrict access to natural resources (e.g., gathering, hunting, fishing, pharmacopeia), limit land availability, and curb economic activities such as agriculture, livestock, and construction. Conversely, they can provide benefits through compensation measures (e.g., local development projects, cash transfers), job creation (e.g., conservation-related employment, tourism), and ecosystem services (e.g., improved water resources, erosion control, fire prevention).~~

~~The impact of conservation on local well-being remains debated~~ (Adams et al. 2004),~~2004). Yet, no study has systematically quantified the population potentially affected by PAs. While some research has examined population density trends around PAs (Joppa, Loarie, and Pimm 2009; Geldmann et al. 2019), none have assessed the demographic implications of PA creation or expansion.~~

~~The proportion of the population directly impacted by PAs is a critical issue for conservation planning, determining the distribution of benefits and burdens across social groups, shaping both local reception and long-term feasibility. This is~~ particularly ~~relevant~~ in low~~-~~ and lower ~~-~~ middle ~~-~~income countries~~,~~ where rural households ~~represent the majority of the population (64% in 2022) and~~ account for a large share of the population and global poverty (Castañeda et al. 2018; ~~). Recent reductions in poverty rates in developing countries have largely resulted from declines in rural poverty (~~Byerlee and Janvry 2008). ~~), underscoring the importance of understanding how conservation policies intersect with rural livelihoods. PA expansion is thus not solely an ecological decision but a political one, affecting policy support from local elites, voters, and decision-makers, who balance conservation goals with social welfare mandates (Mangonnet, Kopas, and Urpelainen 2022). Recognizing the demographic extent of PA-affected populations is therefore crucial, as neglecting their socioeconomic realities risks generating resistance and conflict, undermining conservation effectiveness.~~

The ~~rapid~~ expansion of PAs makes this issue increasingly salient: from~~pressing. From~~ 2000 to 2023, global terrestrial PA coverage expanded by 62%, reaching 17.6% of land area (UNEP-WCMC and IUCN 2024). This growth is set to accelerate with the COP15 commitment to protect 30% of terrestrial land by 2030 ~~(. While crucial for biodiversity, these efforts also raise urgent questions about socioeconomic impacts. As more land becomes protected, more people will be affected, either through restricted resource access or conservation-driven economic opportunities. Quantifying how many people live near PAs is essential to designing policies that balance ecological and human well-being.~~2022).

In the literature assessing the social and economic implications of protected areas, the effects of conservation are consistently framed as local in nature. Yet what constitutes a "local population" is rarely made explicit in demographic terms, even though these definitions implicitly determine the scale to which empirical results apply and can be compared. Two dominant notions of locality recur. One focuses on populations living inside protected areas, directly subject to restrictions and management regimes. The other extends locality to populations living outside but near protected areas, on the premise that social effects attenuate with distance, reflecting spatial limits to resource use and interaction documented in earlier work (West, Igoe, and Brockington 2006).

A 10 km buffer around PA boundaries has emerged as a widely used threshold. It was the inclusion criteria in the widely cited systematic review on of conservation outcomes by Oldekop et al. (2016). It is also pervasive in individual empirical studies. In one of the few global-scale assessments, Naidoo et al. (2019) analyzed conservation wellbeing outcomes across 34 low- and middle-income countries with Demographic and Health Surveys, defining exposure to protected areas as residing at 10 km or less from PAs. Fisher et al. (2024) used Afrobarometer data to assess social wellbeing, explicitly classifying households based

a mis en forme : First Paragraph

on their distance to protected area boundaries, including a 10 km threshold. Among smaller-scale impact evaluations, spatial definitions vary but converge around similar thresholds. Kandel et al. (Kandel et al. 2022) meta-analyze 30 counterfactual studies. Eleven define exposure through in–out comparisons between populations residing inside protected areas and those outside; two of these explicitly exclude populations living within 10 km and 20 km, respectively, from the control group because they may be affected by spillover effects. Ten studies use distance-based or buffer definitions, five of which refer to a 10 km threshold. Nine rely on administrative units that overlap protected areas to varying degrees, reflecting the spatial resolution of available socioeconomic data. Across all these designs, the population sizes implied by exposure definitions are not reported.

This paper addresses this gap by quantifying the demographic scale of conservation exposure definitions. Rather than assessing conservation impacts or making claims about them, it asks a simpler and different question: when studies of PA social impacts define "local populations" spatially, how many people does this actually correspond to, and how has this magnitude evolved since 2000, as protected area systems have expanded? Using globally harmonized PA boundaries and gridded population data, the paper quantifies population counts inside PAs and within a reference buffer commonly used in the literature, and documents how these numbers vary across countries, protected area categories, and time. Providing population orders of magnitude is necessary for interpreting the scope and external validity of estimated conservation social impacts.

Methods

Study Area

The analysis~~study~~ focuses on countries classified by the World Bank as low-income or~~and~~ lower-middle-income economies in 2020~~, based on Gross National Income per capita, adjusted using the Atlas method~~ (Vaggi 2017). ~~We excluded~~ South Sudan and Timor-Leste~~, as they~~ did not exist as sovereign states at the beginning of our analysis period (~~in~~ 2000), so we did not include them in the analysis.

India is excluded because data on its nationally designated protected ~~. The~~ areas are restricted from public access in the WDPA. The WDPA records approximately 990 protected areas for India, but only 90 internationally designated sites (Ramsar wetlands, World Heritage sites, UNESCO-MAB Biosphere Reserves) are publicly available, covering approximately 0.5% of India's land area; data on the remaining 900 nationally designated sites cannot be viewed or downloaded (UNEP-WCMC 2026). Including ~~currently corresponding to~~ these restricted sites, India's terrestrial PA coverage reaches approximately 7.7%. India has stated that the process of making its data publicly accessible is in progress (Wani et al. 2025), and this data gap has been noted in global assessments (Zhang et al. 2025). Because India accounts for approximately 36% of the total LMIC population, including it with incomplete spatial data would mechanically depress aggregate PA coverage and exposure statistics. The analysis therefore covers 75 LMICs~~two countries were removed from the 2000 dataset~~.

a mis en forme : Corps de texte

~~We set a 10 km threshold to define the treatment group, a standard practice for assessing socioeconomic impacts of protected areas in developing countries (Naidoo et al. 2019; Oldekop et al. 2016).~~ The population of interest is defined following spatial exposure concepts commonly used in the literature evaluating the impacts of protected areas. Two reference perimeters are considered: residence inside protected areas, and residence within a 10 km buffer around protected area boundaries. These perimeters are used as illustrative thresholds to quantify the population sizes implied by commonly used definitions of "local populations" in conservation impact studies, not as claims about the spatial extent or magnitude of actual conservation impacts.

~~Buffers did not extend across national borders, consistent with the assumption that populations are restricted from crossing them.~~

Data Sources

National~~The national~~ boundaries were drawn from ~~selected using~~ geoBoundaries version ~~(v.~~6.0.0, a~~), an open,~~ standardized~~,~~ and openly available global ~~regularly updated~~ dataset designed for cross-country comparative analysis (Runfola et al. 2020).

Protected area boundaries, status, and designation year are obtained from the World Database on Protected Areas (~~The~~ WDPA), compiled ~~(February 2025 edition), managed~~ by UNEP-WCMC in collaboration with IUCN~~, is the most comprehensive dataset on protected areas, updated monthly with submissions from national governments, non-governmental organizations, landowners, and local communities~~ (Bingham et al. 2019; UNEP-WCMC and IUCN 2023). We use ~~The status year field records~~ the May 2021 WDPA release, ~~year in~~ which underpinned the final assessment~~a site was formally designated, allowing for historical analyses~~ of Aichi Target 11 under the Convention on Biological Diversity (CBD) Strategic Plan for Biodiversity 2011–2020 (UNEP-WCMC and IUCN 2021). This snapshot reflects the culmination of reporting efforts by CBD member countries for the 2020 milestones, and is therefore more likely to be complete, while limiting the inclusion of post-2020 designations recorded with missing status years~~PA expansion~~.

Population estimates are drawn from two globally harmonized gridded datasets. The Global Human Settlement Layer (GHSL) combines census-based population counts with~~draws from Gridded Population of the World (GPWv4) (Center for International Earth Science Information Network-CIESIN-Columbia University 2018). GHSL improves on this by integrating~~ remote sensing and ancillary~~crowdsourced~~ data to produce gridded estimates at 250 m~~, using regression models to refine 250m~~ resolution ~~estimates~~ (Freire et al. 2016). WorldPop applies a machine-learning approach trained on census data and spatial covariates to generate population estimates at 100 m resolution (Stevens et al. ~~; Schiavina, Freire, and MacManus~~ 2015). Both datasets have~~lt has~~ been evaluated against geolocated census data in contexts where such data are available (Leyk et al. 2019;~~assessed as the most reliable global gridded population dataset by~~ Chen et al. ~~(~~2020; Thomson et al. 2022).~~), and we use~~ GHSL is used~~population estimates~~ as the main~~default~~ reference dataset, while WorldPop is used to assess the robustness of population orders of magnitude~~unless stated otherwise~~.

~~The second-best dataset according to Chen et al. (2020) is Worldpop, which applies a random forest algorithm trained on census data, land cover, infrastructure, elevation, and mobile phone records to produce high-resolution (100m) estimates (Stevens et al. 2015). It supports spatially explicit analyses of human presence near PAs. We use its 2000 and 2020 datasets (WorldPop and CIESIN 2020).~~

Analytical approach

We used Google Earth Engine (GEE), a cloud-based platform for large-scale environmental analysis (Gorelick et al. 2017), to process high-resolution population estimates within and around protected areas across the 75 countries in 2000 and 2020~~. The computation, executed in iterative batches, handled billions of pixels over 100 hours~~.

Protected areas were filtered to retain only designated, established, or inscribed sites~~We extracted protected area boundaries from the WDPA~~, excluding UNESCO-MAB Biosphere Reserves ~~,~~ as is common practice in conservation analyses due to their often limited legal protection (Hanson 2022; Coetzer, Witkowski, and Erasmus 2014) – and purely marine protected areas, since this study focuses on ~~). We retained only~~ terrestrial populations. Protected areas were classified into three groups based on reported IUCN management categories, following Lebergers et al. (2020): strict protected areas (status I to III), non-strict protected areas (status IV to VI), and protected areas whose IUCN category is not recorded in the WDPA. Protected areas were considered as existing by the end~~PAs or the terrestrial portion~~ of the study period if they had a reported designation year less than or equal to 2020 or if the designation year was missing in the May 2021 WDPA release. For analyses of change over time, protected areas with missing designation year were ~~coastal PAs that intersected geoBoundaries and~~ excluded to avoid ambiguity in temporal classification.

For each designation-period class and protected area category,~~purely offshore PAs. We generated~~ binary raster masks were constructed~~PA presence layers at 100m resolution~~ and expanded using a~~created~~ 10 ~~-~~km buffer ~~zones~~ to represent~~assess population~~ proximity-based reference perimeters. Population counts and protected area surface were computed through~~. We applied~~ zonal aggregation within ~~statistics to WorldPop data, aggregating results at the~~ national boundaries. Spatial computations are conducted within national boundaries, and proximity buffers around protected areas are truncated at international borders~~level for small and medium-sized countries (<1,000,000 km^2) and at the subnational level for large ones to manage memory constraints~~.

~~To assess the consistency of GHSL and WorldPop population estimates, we compared values per country and year. Absolute differences were unsuitable means of comparison, as they are influenced by the estimate levels, which vary significantly across countries and depending on whether we consider PAs or surrounding areas. We used the Standardized Mean Difference (SMD), which standardizes the difference by accounting for variations in scale and dispersion. It is computed as:~~

$$\cancel{SMD = \frac{|\bar{X}_1 - \bar{X}_2|}{\sqrt{\frac{\sigma_1^2 + \sigma_2^2}{2}}}}$$

~~where $\bar{X}_1$ and $\bar{X}_1$ are the mean population estimates from respectively WorldPop and GHSL, and σ_1^2 and σ_2^2 are their variances.~~

We conducted the final analyses in R (R Core Team 2023~~), using a coherent framework for data processing (Wickham et al. 2019) and packages optimized for graphical visualization (Wickham 2016; Slowikowski 2023; Wilke 2024) and tabular presentation (Iannone et al. 2023~~). The complete replication package, including the GEE JavaScript code, R code, and output dataset, is available on GitHub (https://github.com/BETSAKA/population_near_protected_areas) and archived on Software Heritage (swh:1:dir:2c8237e16e74d94043e37899aed0de6f262ed420).

Results

Population magnitudes

Restricting to populations living inside protected areas refers to a relatively small population: 61 million people, or 2.5% of the combined population of these 75 countries. Extending the exposure definition to include people living within 10 km of PA boundaries increases the encompassed population roughly fifteenfold, to 902 million (36.6%). For comparison, in 2000 (restricted to PAs with recorded designation years), 17.9 million people lived inside PAs and 301.1 million inside or within 10 km (18.2% of the population).

The decomposition by PA category (Figure 1) shows that populations adjacent to non-strict PAs (IUCN categories IV to VI) substantially exceed those adjacent to strict PAs (IUCN categories I to III): 244 million inside or within 10 km of non-strict PAs, compared with 196 million near strict PAs. An additional 462 million people live near PAs whose IUCN category is not recorded in the WDPA.

To connect these magnitudes to the impact evaluation literature, Table 1 translates common spatial exposure definitions into population orders of magnitude.

~~Table S1 summarizes the extent of PAs and population proximity to PAs in low- and lower-middle-income countries. It presents the percentage of national land within PAs and the percentage of population within 10 km of PAs for the years 2000 and 2020. Due to its large population and relatively low PA coverage, India has an overwhelming impact on the global average for developing countries. To account for this, we provide two totals: one including India and another excluding it.~~

~~Terrestrial PA coverage in developing countries increased from 9.3 % in 2000 (3.4 million km2) to 12.9 % in 2020 (4.7 million km2), while the share of the population living within 10 km of a PA according to GHSL rose from 16.4 % (446.5 million people) to 21.8 % (844.4 million people) over the same period. This global average is largely driven by India, due to its~~

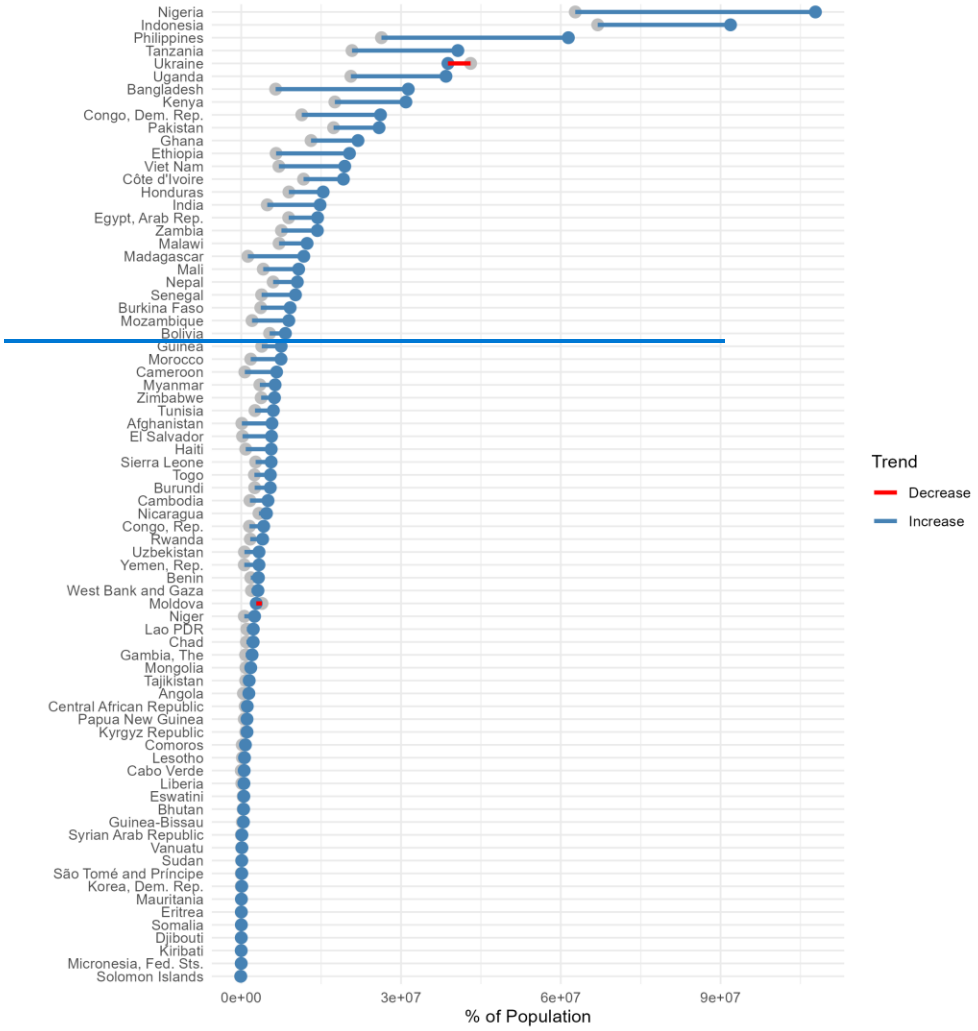
~~demographic size and almost null PA expansion. For the 75 other countries, PA coverage increased from 10.2 % in 2000 (3.4 million km2) to 14.1 % in 2020 (4.7 million km2), while the share of the population living within 10 km of a PA according to GHSL rose from 26.5 % (441.6 million people) to 33.5 % (829.6 million people).~~

~~The expansion of PA coverage was contrasted across countries.Some countries saw substantial PA expansion (e.g., Bhutan, Cambodia), while others experienced minimal growth (e.g., India, Ghana, Côte d'Ivoire). Countries with already high PA coverage in 2000 saw only marginal increases (e.g., Zambia, Tanzania).~~

~~Shifts in Shares of Population Near Protected Areas (2000–2020)~~

~~Figure 1 illustrates country-level changes in the percentage of population residing within 10 km of protected areas between 2000 and 2020. The points represent the population share near PAs in both years, while the connecting lines indicate whether the trend is increasing (blue) or decreasing (red).~~

Figure 1: Change in % Population Near PAs (GHSL 2000-2020)



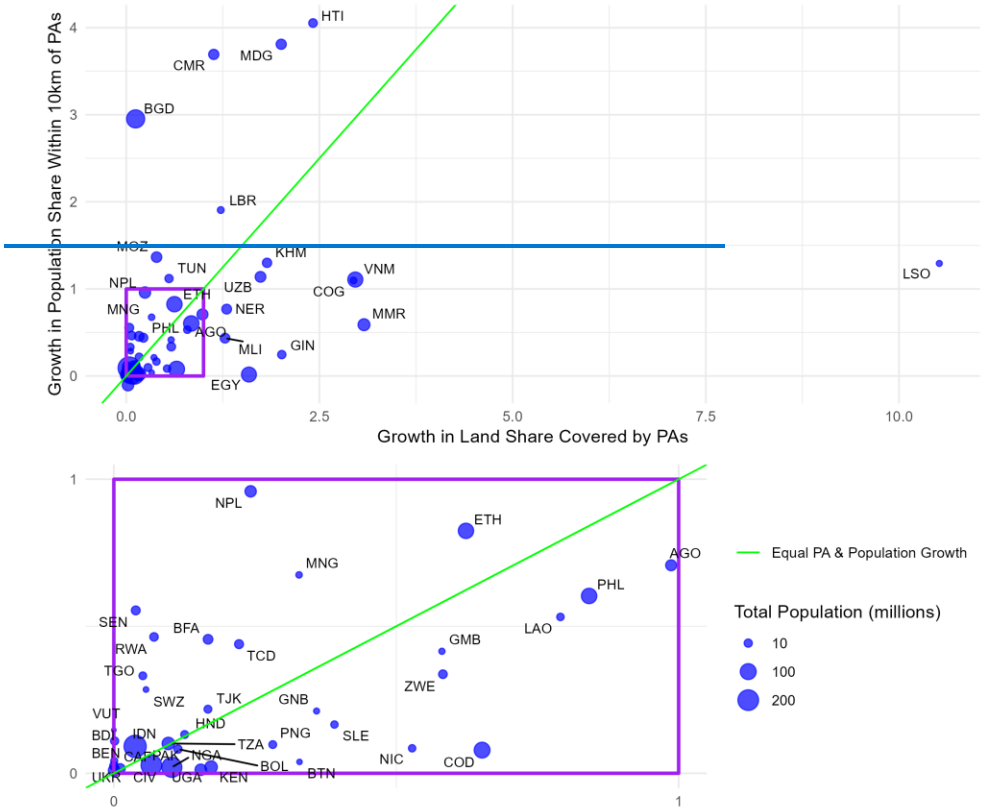
~~Most countries experienced an increase in PA-adjacent populations, with particularly sharp rises in Comoros, Cabo Verde, El Salvador, Tunisia, São Tomé and Príncipe, Haiti, and Madagascar, where PA expansion and population growth intensified proximity. In contrast, Ukraine and Moldova saw slight declines due to urban demographic shifts despite stable PA coverage.~~

~~Growth of PA Coverage and Population near PAs between 2000 and 2020~~

~~Figure 2 presents changes in PA land coverage and the population share living within 10 km of PAs. The diagonal green line represents equal growth in both metrics. Countries above the line experienced faster growth in PA-adjacent populations than in PA coverage.~~

Figure 2: Evolution of PA Land Coverage and Nearby Population Share (GHSL 2000-2020)

Subsample of 58 countries classified as Low- or Low-Middle Income in 2020, with at least 1% PA coverage in 2000



Country codes: AFG: Afghanistan, AGO: Angola, BDI: Burundi, BEN: Benin, BFA: Burkina Faso, BGD: Bangladesh, BOL: Bolivia, BTN: Bhutan, CAF: Central African Republic, CIV: Côte d'Ivoire, CMR: Cameroon, COD: Congo, Dem. Rep., COG: Congo, Rep., COM: Comoros, CPV: Cabo Verde, DJI: Djibouti, EGY: Egypt, Arab Rep., ERI: Eritrea, ETH: Ethiopia, FSM: Micronesia, Fed. Sts., GHA: Ghana, GIN: Guinea, GMB: Gambia, The, GNB: Guinea-Bissau, HND: Honduras, HTI: Haiti, IDN: Indonesia, IND: India, KEN: Kenya, KGZ: Kyrgyz Republic, KHM: Cambodia, KIR: Kiribati, LAO: Lao PDR, LBR: Liberia, LSO: Lesotho, MAR: Morocco, MDA: Moldova, MDG: Madagascar, MLI: Mali, MMR: Myanmar, MNG: Mongolia, MOZ: Mozambique MRT: Mauritania, MWI: Malawi, NER: Niger, NGA: Nigeria, NIC: Nicaragua, NPL: Nepal, PAK: Pakistan, PHL: Philippines, PNG: Papua New Guinea, PRK: Korea, Dem. Rep., PSE: West Bank and Gaza, RWA: Rwanda, SDN: Sudan, SEN: Senegal, SLB: Solomon Islands, SLE: Sierra Leone, SLV: El Salvador, SOM: Somalia, STP: São Tomé and Príncipe, SWZ: Eswatini, SYR: Syrian Arab Republic, TCD: Chad, TGO: Togo, TJK: Tajikistan, TUN: Tunisia, TZA: Tanzania, UGA: Uganda, UKR: Ukraine, UZB: Uzbekistan, VNM: Viet Nam, VUT: Vanuatu, YEM: Yemen, Rep., ZMB: Zambia, ZWE: Zimbabwe

~~Most countries fall below the diagonal, where PA spatial expansion outpaced the growth of demographic share of adjacent populations. Lesotho in particular, exhibit large PA expansions with relatively small shifts in population share, indicating that new PAs were primarily established in low-density areas. This reflects the classical trend of PAs to be typically established "high and far", where pressure for land conversion is lower (Joppa and Pfaff 2009).~~

~~In contrast, countries in the top left quarter of Figure 2 experienced a much larger increase in the percentage of their population near PAs than in PA coverage, indicating that new PAs or expansions occurred in relatively high-density areas. Such a pattern is expected in countries with predominantly rural populations, such as Madagascar, Bangladesh, and Mozambique, where PA establishment is more likely to affect human settlements In other cases, such as Haiti, Cameroon, and Liberia, the trend appears to result from deliberate conservation choices to designate PAs in more densely inhabited regions, rather than in remote areas. The zoomed-in portion highlights countries with smaller relative changes, where PA coverage and nearby population increased less than twofold between 2000 and 2020.~~

~~Share of PA Coverage and Population Near PAs in 2020~~

~~Figures 3 displays the relationship between the percentage of national land designated as protected areas and the percentage of the population residing within 10 km of these areas in 2020. Each point represents a country, with country codes labeled on the graph and a corresponding key provided below. The size of the points reflects the total national population.~~

Figure 3: PA Coverage vs Population Near PAs (2020)

Country codes: AFG: Afghanistan, AGO: Angola, BDI: Burundi, BEN: Benin, BFA: Burkina Faso, BGD: Bangladesh, BOL: Bolivia, BTN: Bhutan, CAF: Central African Republic, CIV: Côte d'Ivoire, CMR: Cameroon, COD: Congo, Dem. Rep., COG: Congo, Rep., COM: Comoros, CPV: Cabo Verde, DJI: Djibouti, EGY: Egypt, Arab Rep., ERI: Eritrea, ETH: Ethiopia, FSM: Micronesia, Fed. Sts., GHA: Ghana, GIN: Guinea, GMB: Gambia, The, GNB: Guinea-Bissau, HND: Honduras, HTI: Haiti, IDN: Indonesia, IND: India, KEN: Kenya, KGZ: Kyrgyz Republic, KHM: Cambodia, KIR: Kiribati, LAO: Lao PDR, LBR: Liberia, LSO: Lesotho, MAR: Morocco, MDA: Moldova, MDG: Madagascar, MLI: Mali, MMR: Myanmar, MNG: Mongolia, MOZ: Mozambique MRT: Mauritania, MWI: Malawi, NER: Niger, NGA: Nigeria, NIC: Nicaragua, NPL: Nepal, PAK: Pakistan, PHL: Philippines, PNG: Papua New Guinea, PRK: Korea, Dem. Rep., PSE: West Bank and Gaza, RWA: Rwanda, SDN: Sudan, SEN: Senegal, SLB: Solomon Islands, SLE: Sierra Leone, SLV: El Salvador, SOM: Somalia, STP: São Tomé and Príncipe, SWZ: Eswatini, SYR: Syrian Arab Republic, TCD: Chad, TGO: Togo, TJK: Tajikistan, TUN: Tunisia, TZA: Tanzania, UGA: Uganda, UKR: Ukraine, UZB: Uzbekistan, VNM: Viet Nam, VUT: Vanuatu, YEM: Yemen, Rep., ZMB: Zambia, ZWE: Zimbabwe

~~Figure 3 highlights considerable variation across countries in both PA coverage and the proportion of people living near these areas. Some countries, such as Bhutan and Tanzania, have extensive PA networks. A cluster of countries with low PA coverage and limited nearby population presence can also be observed. Notably India falls in this section, and it has such a large population that it weighs heavily on the global averages drawn from these national statistics.~~

~~Consistency between sources~~

~~Table 1 presents the SMDs between GHSL and WorldPop estimates for populations residing within and near protected areas in 2000 and 2020, with higher values indicating greater discrepancies.~~

Table 1: Implied population magnitudes by evaluation design~~Standard Mean Differences Between GHSL and Worldpop Estimates~~

All PAs in 2020 (incl. unknown designation year), 75 LMICs (excl. India), GHSL population estimates

Evaluation design (exposure definition)	~~2000~~Population (millions)	% of LMIC population~~2020~~
In-out comparison (inside PAs only)	60.7	2.5
Inside PAs or~~**Population**~~ within 10 km~~**PAs**~~	902.1~~0.29~~	36.6~~0.20~~
Strict PAs only (IUCN Ia-III): inside or~~**Population**~~ within 10 km~~**10km of PAs**~~	196.1~~0.09~~	8.0~~.08~~
Non-strict PAs only (IUCN IV-VI): inside or within 10 km	243.5	9.9
Unknown category PAs: inside or within 10 km	462.5	18.8

Source: WDPA May 2021, GHSL 2020. 'Within 10 km' includes populations inside PAs.

a mis en forme : Police :+Corps (Aptos), 12 pt

a mis en forme : Gauche

a mis en forme : Police :Non Gras

a mis en forme : Gauche

a mis en forme : Police :Non Gras

a mis en forme : Gauche

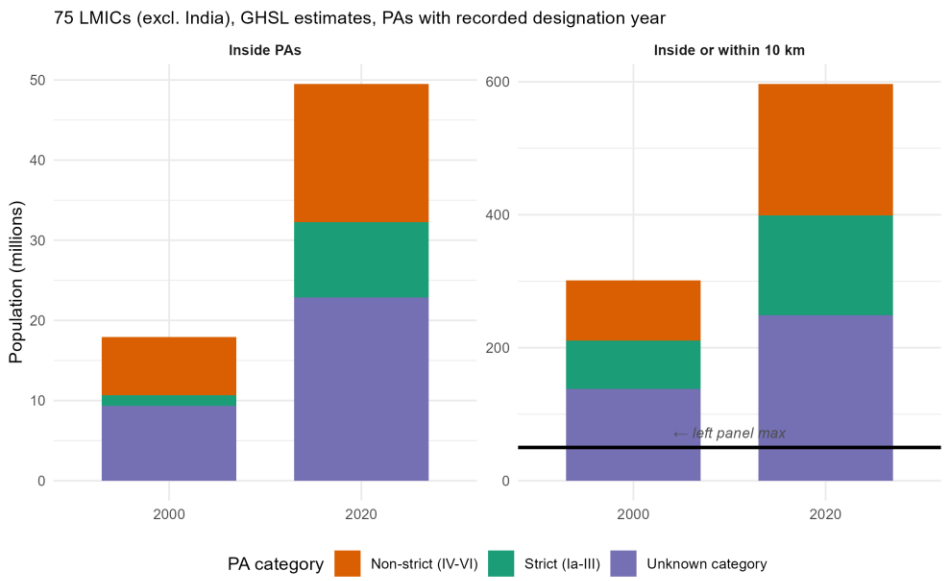
An in-out comparison – where "treated" populations are those residing inside PAs – implies a study population of approximately 61 million people (2.5% of the population of these 75 countries). When the spatial definition extends to populations within 10 km, the implied population rises to 902 million (36.6%). Restricting to strict PAs only yields 196 million, while non-strict PAs contribute 244 million. These numbers do not appear in the studies reviewed by Kandel et al. (2022), yet they define the implicit scope to which estimated impacts apply.

Change over time by PA category

Comparing 2020 with 2000 is subject to an important caveat: designation years are missing for 690 out of 3,154 PAs in the WDPA (across the 75 LMICs studied), so the set of PAs that can be confirmed as existing before 2001 is a subset of those known to exist by 2020. The 2000-2020 comparison is therefore restricted to PAs with recorded designation years and should be interpreted as indicative of broad trends rather than a precise measure of change.

With this caveat in mind, ~~There is no universal interpretation of SMD values for comparing different population datasets. Cohen's rule of thumb suggests an SMD below 0.2 represents a negligible difference, while Austin proposes a more conservative threshold of 0.1 for comparability (Cohen 1988; Austin 2009). Applying this principle, GHSL and WorldPop differ significantly for PA-internal population estimates but align well for populations within 10 km of PAs. GHSL is considered more reliable for high- and low-density settings (Chen et al. 2020).~~

Figure 1 shows how PA-adjacent populations evolved between 2000 and 2020 across PA categories, separately for populations inside PAs and within the 10 km buffer.



Growth in exposed populations occurred across all three PA categories. The net change (2020 minus 2000) for the 75 LMICs combined (Figure S1, Supplementary Materials) shows that the population inside or within 10 km of non-strict PAs grew by approximately 108 million, that inside or within 10 km of PAs with unrecorded IUCN category by 111 million, and that near strict PAs by 77 million.

This overall increase reflects two distinct mechanisms: population growth near PAs that already existed by 2000, and the designation of new PAs between 2000 and 2020. Using national population growth rates to approximate what 2000-era PA footprints would have contained under 2020 population levels, we estimate that roughly 53% of the increase is attributable to demographic growth around pre-existing PAs, and 47% to PA network expansion. The two mechanisms contribute in approximately equal measure at the global LMIC level, though their relative importance varies substantially across countries.

Uncertainty from missing designation years

The WDPA records a designation year for each protected area, but this field is not reported for 690 of the 3,154 PAs in our 75 LMICs (see Supplementary Materials, Section A). These PAs are known to exist, since they appear in the May 2021 WDPA release, but they cannot be unambiguously assigned to a specific designation period. Figure S2 (Supplementary Materials) shows, for each country, the gap between two counts: one restricted to PAs whose designation year is explicitly recorded as falling on or before 2020, and one that also includes PAs with missing designation year. The gap represents the uncertainty attributable to incomplete temporal metadata in the WDPA.

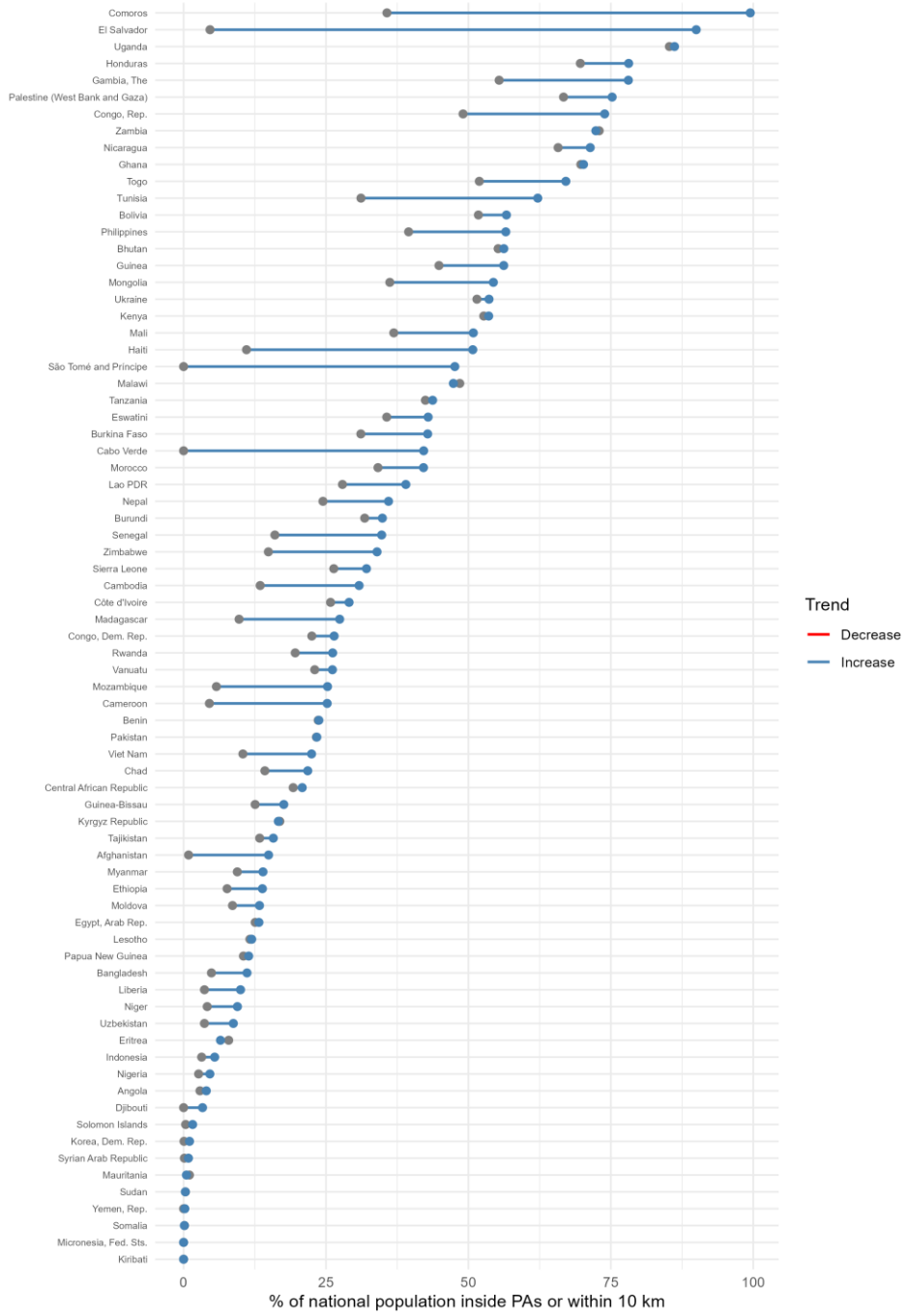
For 54 of the 75 countries, the gap is below 5 percentage points, indicating that the large majority of PAs have recorded designation years. However, in several countries – notably those where IUCN categories are missing for large protected areas – the gap exceeds 10 percentage points. For these countries, population figures are sensitive to how missing designation years are treated.

Cross-country variation

Figure 2 displays, for each country, the percentage of the national population residing inside PAs or within 10 km, in 2000 and 2020. The wide cross-country variation – from less than 1% to over 90% – underscores that aggregate statistics mask considerable heterogeneity. In 66 of the 75 countries, the share strictly increased between ~~4 provides a disaggregated comparison of GHSL~~ 2000 and 2020 (blue segments), consistent with PA expansion. Two island states (Micronesia and Kiribati) had no PAs in either period. Seven countries experienced marginal declines (all below 1.5 percentage points), most notably Eritrea, Malawi, and Zambia, where stable PA coverage combined with population growth in non-PA areas slightly reduced the relative share of PA-adjacent populations. Full country-level detail is provided in Table S1 (Supplementary Materials).

Figure 2: Change in % population near PAs (2000-2020)

75 LMICs (excl. India), GHSL estimates, PAs with recorded designation year.



Robustness: GHSL vs WorldPop

Figure 3 compares GHSL and WorldPop estimates of the percentage of national population inside PAs and within 10 km, for all PAs present in the 2020 WDPA release. Points cluster around the 1:1 line, indicating broad agreement between the two datasets at the country level.

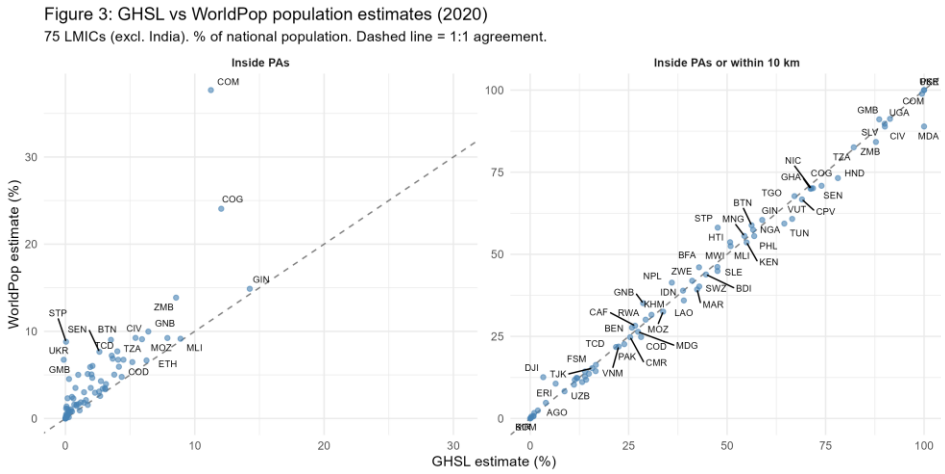
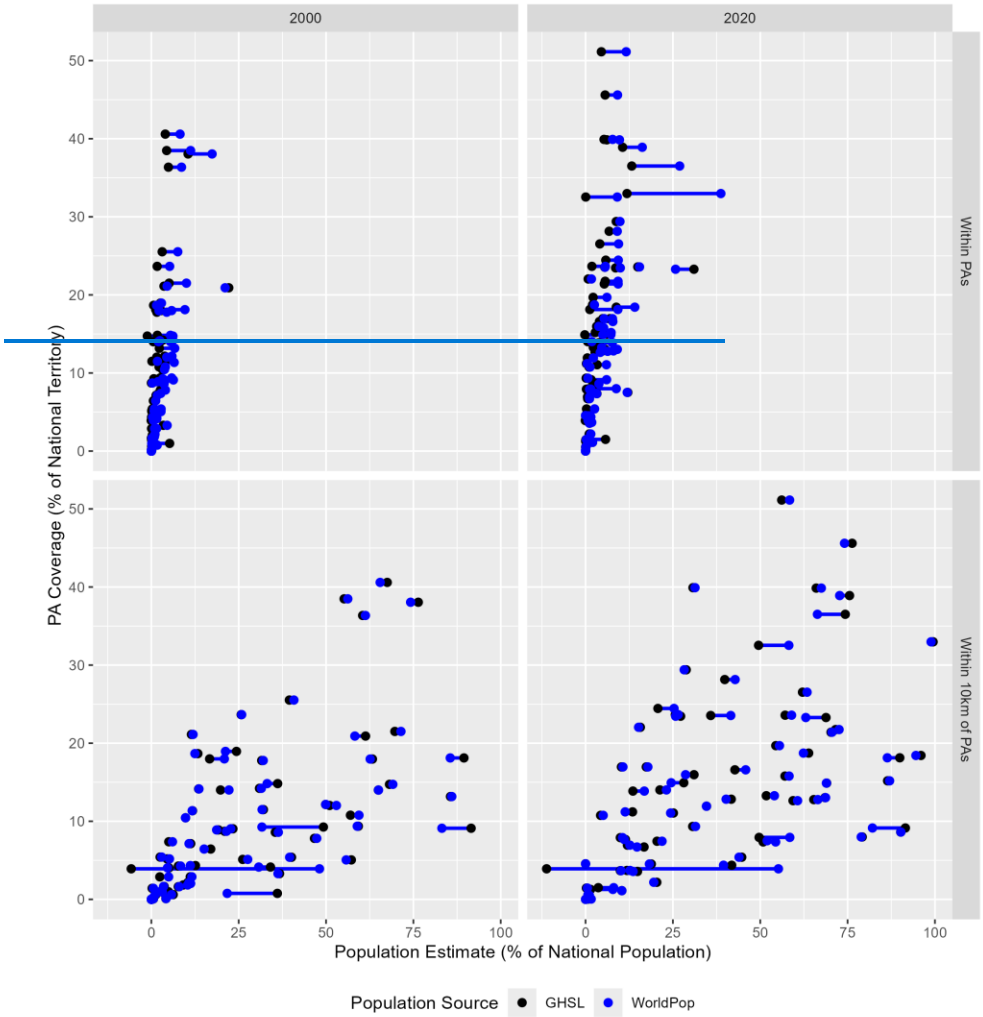


Figure 1

Discrepancies are most pronounced for populations inside PAs, where the very small absolute population counts amplify relative differences between the two datasets' spatial allocation models. For the broader 10 km buffer perimeter, agreement is substantially stronger. Table S2 (Supplementary Materials) lists the countries where the absolute difference between~~. Each segment represents the difference between the two sources for a given country, with black and blue points denoting~~ GHSL and WorldPop exceeds 5 percentage points and the relative difference (computed as the absolute difference divided by the GHSL estimate) exceeds 10%. These discrepancies do not alter the order-of-magnitude conclusions: both datasets indicate hundreds of millions of people residing near protected areas in LMICs~~, respectively~~.

Figure 4: Comparison of Population Estimates from GHSL and WorldPop



~~For most countries, GHSL and WorldPop estimates align closely, with overlapping black and blue dots on Figure 4. However, some countries exhibit large discrepancies. Table S1 in the appendix details the 15 countries where GHSL and WorldPop estimates differ by more than 5 percentage points and by more than 10%. Notably, WorldPop estimates significantly exceed GHSL in countries such as Comoros, the Republic of Congo, and São Tomé and Príncipe, while GHSL reports higher population shares near PAs than WorldPop in countries such as Ukraine and Zambia. These variations may stem from differences in population allocation methodologies or underlying census data.~~

Discussion

Demographic scale and the impact evaluation literature

The central finding of this paper is that the populations encompassed by standard spatial exposure definitions in PA impact evaluations are large – on the order of several hundred million people. Among PAs with recorded designation years, these populations are substantially larger in 2020 than in 2000, driven both by PA expansion itself and by population increase in areas where PAs were already present.

These population orders of magnitude are absent from the impact evaluation literature - whether in meta-analyses such as Kandel et al. (2022) or in large-scale assessments such as Naidoo et al. (2019) and Fisher et al. (2024). Yet they define the scope to which estimated impacts implicitly apply. When a study estimates the welfare effect of living within 10 km of a PA in a given country, and that effect is used to inform policy discussions about PA expansion, it is relevant to know that the population meeting this exposure criterion across LMICs numbers in the hundreds of millions - and that this population is growing.

This matters for at least three reasons. First, it sets a demographic frame for external validity: a treatment effect estimated on a sample of a few thousand households in a handful of countries implicitly speaks to a reference population that is orders of magnitude larger. Second, it reveals compositional heterogeneity: the "local population" implied by a 10 km buffer near a strict national park in a sparsely populated area is demographically very different from that implied by the same buffer near a densely inhabited community conservation area. Third, the heterogeneity of findings in the impact evaluation literature may itself partly reflect the heterogeneity of the populations being studied – populations that differ not only across countries but across PA management categories, as our decomposition shows.

PA categories matter for interpreting impacts

The decomposition by IUCN management category reveals that populations adjacent to non-strict PAs (IUCN IV-VI) substantially exceed those adjacent to strict protection regimes (IUCN I-III). This pattern held in both 2000 and 2020.

This has direct relevance for interpreting impact evaluation results. The term "protected area" encompasses management regimes with vastly different implications for local populations – from national parks that prohibit entry to community conservation areas that allow resource extraction – yet comparisons of impacts across management categories remain rare (Kandel et al. 2022; Oldekop et al. 2016). Our results show that the demographic weight of non-strict PAs exceeds that of strictly protected areas in LMICs. Studies that treat all PAs as a homogeneous treatment condition may therefore conflate very different exposure regimes. PAs whose IUCN category is not recorded in the WDPA represent an additional source of ambiguity: for these areas, even the management regime is unknown.

Assessing what these different management regimes imply for local welfare is the domain of impact evaluation studies, not the purpose of this paper. What we document is that the population living near PAs is concentrated around non-strict protection regimes.

Data limitations

Our estimates rely on globally harmonized gridded population datasets. These datasets have known limitations. GHSL and WorldPop are constructed from census data combined with remote sensing and machine learning, and their accuracy varies across settings. Chen et al. (2020) evaluated four global gridded datasets and found that GHSL exhibited the closest correspondence to administrative census totals, while WorldPop showed larger deviations from census totals but high spatial consistency with other datasets. Leyk et al. (2019) and Thomson et al. (2022) provide additional assessments of WorldPop's cell-level accuracy. We use GHSL as our primary dataset and WorldPop as a robustness check. The concordance between the two (Figure 3) supports the conclusion that population orders of magnitude are robust to the choice of dataset, even if point estimates for individual countries may differ.

The exclusion of India deserves further comment. India accounts for approximately 36% of the total LMIC population, and its actual PA network – 900 nationally designated sites, for a terrestrial coverage of approximately 7.7% (UNEP-WCMC 2026) – is substantial. However, India has restricted public access to these data in the WDPA, making only internationally designated sites (Ramsar wetlands, World Heritage sites) available, covering less than 1% of Indian territory. This data restriction has been documented: Wani et al. (2025) note that India has stated the process of making its data publicly accessible is in progress, and Zhang et al. (2025) acknowledge that the absence of publicly available Indian PA data may underestimate conservation extent in global analyses. Including India with incomplete spatial data would mechanically depress aggregate exposure statistics, producing artefactual rather than substantive results. We therefore exclude India entirely. This means our estimates do not cover the world's most populous lower-middle-income country, which is a significant limitation. If India's PA data were publicly available, the total number of people estimated to live near protected areas would increase substantially. At the same time, the share of the population concerned would likely decrease, given India's comparatively low protected area coverage.

The treatment of missing designation years introduces temporal ambiguity (see ~~While this study does not offer new theoretical claims or seek to estimate causal effects of protected area expansion, it addresses an empirical gap with significant implications for both research and policy. By quantifying the number of people living in or near protected areas across the developing world, we provide a baseline understanding of the outreach of conservation policies on neighbourhood communities. This "mere description" (Gerring 2012) is essential for framing future work on the socio-economic impacts of protected areas and for informing policy debates around equitable and effective conservation.~~

~~The demographic dynamics around protected areas have been a subject of debate in the literature. Wittemyer et al. (2008) reported accelerated population growth near existing PAs, suggesting an attraction effect, while Joppa, Loarie, and Pimm (2009) argued that such cases~~

~~were isolated and that population growth near PAs was not disproportionately high overall. Our analysis aligns with the latter, as we find no evidence of increased neighboring population growth in areas where PA coverage has remained stable. Instead, our results indicate that the observed rise in populations near PAs in some countries is primarily driven by the expansion of PA coverage itself. In the cases where population growth outpaces PA expansion, this can be attributed to the selection of denser areas for new PAs. Additionally, a geometric effect may also play a role: as smaller and more irregularly shaped PAs are added to enhance connectivity (Saura et al. 2019) or accommodate political and ecological constraints, the relative extent of surrounding buffer zones increases compared to the total protected area (Schauman et al. 2024).~~

~~Our estimation of the share of people living near protected areas is derived from GHSL and WorldPop and depends on their reliability. Several studies have attempted to assess Worldpop's accuracy, but they all face a key challenge: census data are only available at the administrative unit level, and no alternative dataset provides fine-grained "ground truth" population estimates at low resolution. Consequently, most assessments compare gridded datasets aggregated to administrative units with official census counts or evaluate the spatial consistency across gridded population datasets. The most comprehensive study is Chen et al. (2020), which evaluated four global gridded datasets – WorldPop, HYDE (Goldewijk et al. 2017), GPWv4, and GHSL – across four countries with diverse population distribution patterns. GPWv4 and GHSL were closer to census totals, while WorldPop had larger deviations but the highest spatial consistency with other datasets, suggesting its population allocation model aligns well with relative distribution patterns.~~

~~In principle, household surveys such as the Demographic and Health Surveys (DHS) could complement or validate these gridded estimates. naidoo2019 used DHS data from 34 developing countries (2001–2011) to compare households near and far from PAs in assessing conservation impacts. While their objective was not to estimate the proportion of the population near PAs, their supplementary materials report the number of surveyed households located in proximity to PAs. Direct comparisons between our estimates and household survey data are not straightforward, but a rough approximation using data extracted from Naidoo et al. (2019)'s appendix provides a useful reference point. Summing the total number of surveyed households and the number located within 10 km of a PA, we estimate that approximately 22% of households in the DHS sample reside within 10 km buffers, which is broadly in line with our estimates. However, several factors hinder direct comparison. 1. DHS estimates require inverse probability weighting to be population-representative, which cannot be applied using Naidoo et al. (2019)'s appendix data. 2. DHS GPS coordinates represent the centroid of the enumeration area, not individual household locations. 3. Additionally, DHS relies on a two-stage cluster sampling approach, meaning that the survey covers only a limited number of sampled locations (typically a few hundred per country), reducing spatial representativity at a fine scale. 4. DHS also reports household counts, not individual population estimates, meaning that differences in household size could affect comparability. 5. DHS geolocations are intentionally displaced, with rural clusters randomly shifted by up to 10 km, introducing uncertainty in proximity assessments (Skiles et al. 2013). Processing to the raw DHS data would allow to tackle the issues 1., 4.,~~

~~and update this analysis with more surveys. However, obtaining DHS data also involves a vetting procedure with a USAID contractor and, at the time of writing, the foreign assistance programs have been indefinitely suspended and it is unknown when or if it will be resumed.~~

Appendix A). In our 2000-2020 comparison, we restrict attention to PAs with a recorded designation year, which means we are comparing a subset of PAs known to exist by 2000 with a different subset known to exist by 2020. This comparison is informative about orders of magnitude but should not be read as a precise estimate of temporal change, since some PAs with missing designation years may in fact have been established before 2000. Figure S2 (Supplementary Materials) makes the resulting uncertainty band visible at the country level.

Scope and interpretive boundaries

This paper quantifies population magnitudes. It does not estimate socio-economic impacts, welfare effects, or causal relationships. No claim is made - explicitly or implicitly - about whether living near a protected area makes people better or worse off. Such questions require impact evaluation designs with appropriate identification strategies, which are beyond the scope of this work.

We also do not rank conservation policies or PA categories in terms of desirability. The decomposition by management category is intended to describe the demographic composition of PA-adjacent populations, not to evaluate the relative merits of different protection regimes.

The paper does not claim that spatial proximity constitutes “being affected by” a protected area. The 10 km buffer is used because it is a common reference perimeter in the impact evaluation literature, not because it represents a known threshold for social or ecological effects. The actual spatial extent of PA-related effects on livelihoods is an empirical question that varies by context and cannot be resolved by our data.

A related limitation concerns population differentiation. Our estimates treat all inhabitants within a given perimeter as demographically equivalent. In practice, populations near protected areas are heterogeneous in their dependence on natural resources, their legal status, and their vulnerability to land-use restrictions. Indigenous peoples and traditional communities - estimated at 476 million people worldwide across more than 90 countries (United Nations Department of Economic and Social Affairs 2021) - are disproportionately represented among populations living within or adjacent to protected areas, and often bear the most direct consequences of conservation management (Schleicher et al. 2019). Our gridded population data cannot identify these groups. The aggregate population magnitudes we report therefore encompass communities with very different relationships to protected areas, from urban residents who may experience diffuse ecosystem service benefits to indigenous groups whose livelihoods and territorial rights may be directly affected by PA governance. Disaggregating these populations remains an important challenge for the field.

Conclusion

Across 75 low- and lower-middle-income countries, protected areas are adjacent to populations on the order of several hundred million people. Restricting to PA interiors captures a relatively small population; extending to 10 km increases the encompassed population roughly fifteenfold. Among PAs with recorded designation years, these figures are substantially larger in 2020 than in 2000, reflecting both PA expansion and population increase – though incomplete temporal metadata in the WDPA means that the precise magnitude of this change cannot be established with certainty. Populations near non-strict PAs (IUCN IV-VI) substantially exceed those near strict protection regimes (IUCN I-III). India, the largest LMIC by population, had to be excluded because its national PA data are absent from the WDPA; if included, global counts of people living near protected areas would likely increase in absolute terms, though the corresponding population share might decrease.

These facts are relevant for how conservation impact evaluation results are interpreted, compared, and generalized. They do not tell us what those impacts are. But they establish the demographic scale that spatial exposure definitions imply – a scale that is almost never reported in the studies that use them.

Bibliography

~~Our analysis provides an assessment of the population living within or in the immediate surroundings of protected areas following their expansion in low- and lower-middle-income countries. Between 2000 and 2020, PA coverage in these countries grew from 9.3% to 12.9%, while the share of people living within 10 km of a PA increased from 16.4% to 21.8%. This global trend is strongly influenced by India, where limited PA expansion and a large population shape the overall figures. Excluding India, PA coverage rose from 10.2% to 14.1%, and the proportion of the population near PAs increased from 26.5% to 33.5%. These figures highlight the extent to which conservation policies affect human populations and the need to address their socio-economic implications. Balancing ecological goals with the well-being and rights of nearby communities will become ever more critical as governments pursue accelerated PA expansion to meet global 30% targets. Policymakers should explicitly integrate poverty alleviation and rural livelihood support measures into conservation frameworks, for instance through participatory governance, compensation mechanisms, and targeted rural development investments around protected areas.~~

~~Although our data and method have inherent limitations, it relies on the most consistent and scalable approach currently available for estimating demographic exposure to conservation policies. Future improvements in population mapping and survey integration could refine these estimates, but we believe that our findings offer a useful first step in quantifying the human dimensions of PA expansion at a global level.~~

~~References~~

Adams, William. M., Ros Aveling, Dan Brockington, Barney Dickson, Jo Elliott, Jon Hutton, Dilys Roe, Bhaskar Vira, and William Wolmer. 2004. "Biodiversity Conservation and the Eradication of Poverty." *Science* 306 (5699): 1146–49. https://doi.org/10.1126/science.1097920.

~~Austin, Peter C. 2009. "Balance Diagnostics for Comparing the Distribution of Baseline Covariates Between Treatment Groups in Propensity-Score Matched Samples." *Statistics in Medicine* 28 (25): 3083–3107.~~

Bingham, Heather C., Diego Juffe Bignoli, Edward Lewis, Brian MacSharry, Neil D. Burgess, Piero Visconti, Marine Deguignet, Murielle Misrachi, Matt Walpole, and Jessica L. Stewart. 2019. "Sixty Years of Tracking Conservation Progress Using the World Database on Protected Areas." *Nature Ecology & Evolution* 3 (5): 737743. https://idp.nature.com/authorize/casa?redirect_uri=https://www.nature.com/articles/s41559-019-0869-3&casa_token=Zmluq7JmhxUAAAAA:Id6cAJrCDpmtydpiga8LlmkB5wb9a1O_bwCGwCeA1gz314-F6VT5ekCOeVMJo_Zrpbh_uuoXf0fwD3Txjg.

Byerlee, Derek, and Alain de Janvry, eds. 2008. *World Developpement Report 2008: Agriculture for Development*. Washington DC: World Bank.

Castañeda, Andrés, Dung Doan, David Newhouse, Minh Cong Nguyen, Hiroki Uematsu, and João Pedro Azevedo. 2018. "A New Profile of the Global Poor." *World Development* 101 (January): 250–67. https://doi.org/10.1016/j.worlddev.2017.08.002.

~~Center for International Earth Science Information Network-CIESIN-Columbia University. 2018. "Gridded Population of the World, Version 4 (GPWv4): Population Density, Revision 11." Palisades, NY, USA: NASA Socioeconomic Data; Applications Center (SEDAC). https://sedac.ciesin.columbia.edu/data/set/gpw-v4-population-density-rev11.~~

Chen, Ruxia, Huimin Yan, Fang Liu, Wenpeng Du, and Yanzhao Yang. 2020. "Multiple Global Population Datasets: Differences and Spatial Distribution Characteristics." *ISPRS International Journal of Geo-Information* 9 (11): 637. https://doi.org/10.3390/ijgi9110637.

Coetzer, Kaera L., Edward T. F. Witkowski, and Barend F. N. Erasmus. 2014. "Reviewing Biosphere Reserves Globally: Effective Conservation Action or Bureaucratic Label?" *Biological Reviews* 89 (1): 82–104. https://doi.org/10.1111/brv.12044.

Convention on Biological Diversity. 2022. "Kunming-Montreal Global Biodiversity Framework." In *Convention Biol. Divers. Kunming-Montreal Glob. Biodivers. Framew.(cbd. Int)*. Vol. CBD/COP/15/L.25. https://seea.un.org/sites/seea.un.org/files/unsc_presentation_jillian_campbell_0.pdf.

Fisher, Joshua, Summer Allen, Greg Yetman, and Linda Pistolesi.~~Cohen, Jacob. 1988. *Statistical Power Analysis for the Behavioral Sciences*. routledge.~~

2024. "Assessing the Influence of Landscape Conservation and Protected Areas on Social Wellbeing Using Random Forest Machine Learning." *Scientific Reports* 14 (1): 11357. https://doi.org/10.1038/s41598-024-61924-4.

Freire, Sergio, Kytt MacManus, Martino Pesaresi, Erin Doxsey-Whitfield, and Jane Mills. 2016. "Development of New Open and Free Multi-Temporal Global Population Grids at 250 m Resolution." *Population* 250: 33.

~~Geldmann, Jonas, Andrea Manica, Neil D. Burgess, Lauren Coad, and Andrew Balmford. 2019. "A Global-Level Assessment of the Effectiveness of Protected Areas at Resisting Anthropogenic Pressures." *Proceedings of the National Academy of Sciences* 116 (46): 23209–15. https://doi.org/10.1073/pnas.1908221116.~~

~~Gerring, John. 2012. "Mere Description." *British Journal of Political Science* 42 (4): 721746.~~

~~Goldewijk, K. K., A. Beusen, J. Doelman, and E. Stehfest. 2017. "Anthropogenic Land Use Estimates for the Holocene–HYDE 3.2." *Earth System Science Data* 9: 927–53. https://doi.org/10.5194/essd-9-927-2017.~~

Gorelick, Noel, Matt Hancher, Mike Dixon, Simon Ilyushchenko, David Thau, and Rebecca Moore. 2017. "Google Earth Engine: Planetary-Scale Geospatial Analysis for Everyone." *Remote Sensing of Environment* 202: 1827. https://www.sciencedirect.com/science/article/pii/S0034425717302900.

Hanson, Jeffrey O. 2022. "Wdpar: Interface to the World Database on Protected Areas." *Journal of Open Source Software* 7 (78): 4594. https://doi.org/10.21105/joss.04594.

~~Iannone, Richard, Joe Cheng, Barret Schloerke, Ellis Hughes, Alexandra Lauer, and JooYoung Seo. 2023. "Gt: Easily Create Presentation-Ready Display Tables." https://CRAN.R-project.org/package=gt.~~

~~Joppa, Lucas N., Scott R. Loarie, and Stuart L. Pimm. 2009. "On Population Growth Near Protected Areas." *PLOS ONE* 4 (1): e4279. https://doi.org/10.1371/journal.pone.0004279.~~

~~Joppa, Lucas N., and Alexander Pfaff. 2009. "High and Far: Biases in the Location of Protected Areas." *PLOS ONE* 4 (12): e8273. https://doi.org/10.1371/journal.pone.0008273.~~

Kandel, Pratikshya, Ram Pandit, Benedict White, and Maksym Polyakov. 2022. "Do Protected Areas Increase Household Income? Evidence from a Meta-Analysis." *World Development* 159 (November): 106024. https://doi.org/10.1016/j.worlddev.2022.106024.

Leberger, Roxanne, Isabel M. D. Rosa, Carlos A. Guerra, Florian Wolf, and Henrique M. Pereira. 2020. "Global Patterns of Forest Loss Across IUCN Categories of Protected Areas." *Biological Conservation* 241 (January): 108299. https://doi.org/10.1016/j.biocon.2019.108299.

Leyk, Stefan, Andrea E. Gaughan, Susana B. Adamo, Alex De Sherbinin, Deborah Balk, Sergio Freire, Amy Rose, Forrest R. Stevens, Brian Blankespoor, and Charlie

Frye~~Mangonnet, Jorge, Jacob Kopas, and Johannes Urpelainen. 2022. "Playing Politics with Environmental Protection: The Political Economy of Designating Protected Areas." *The Journal of Politics* 84 (3): 1453–68. https://doi.org/10.1086/718978.~~

. 2019. "The Spatial Allocation of Population: A Review of Large-Scale Gridded Population Data Products and Their Fitness for Use." *Earth System Science Data* 11 (3): 1385–1409. https://essd.copernicus.org/articles/11/1385/2019/essd-11-1385-2019.html.

Maxwell, Sean L., Victor Cazalis, Nigel Dudley, Michael Hoffmann, Ana SL Rodrigues, Sue Stolton, Piero Visconti, Stephen Woodley, Naomi Kingston, and Edward Lewis. 2020. "Area-Based Conservation in the Twenty-First Century." *Nature* 586 (7828): 217227.

Naidoo, R., D. Gerkey, D. Hole, A. Pfaff, A. M. Ellis, C. D. Golden, D. Herrera, et al. 2019. "Evaluating the Impacts of Protected Areas on Human Well-Being Across the Developing World." *Science Advances* 5 (4): eaav3006. https://doi.org/10.1126/sciadv.aav3006.

Oldekop, Johan A., George Holmes, W. Edwin Harris, and Karl L. Evans. 2016. "A Global Assessment of the Social and Conservation Outcomes of Protected Areas." *Conservation Biology* 30 (1): 133141.

R Core Team. 2023. *R: A Language and Environment for Statistical Computing*. Vienna, Austria: R Foundation for Statistical Computing. https://www.R-project.org/.

Runfola, Daniel, Austin Anderson, Heather Baier, Matt Crittenden, Elizabeth Dowker, Sydney Fuhrig, Seth Goodman, et al. 2020. "geoBoundaries: A Global Database of Political Administrative Boundaries." *PLOS ONE* 15 (4): e0231866. https://doi.org/10.1371/journal.pone.0231866.

~~Saura, Santiago, Bastian Bertzky, Lucy Bastin, Luca Battistella, Andrea Mandrici, and Grégoire Dubois. 2019. "Global Trends in Protected Area Connectivity from 2010 to 2018." *Biological Conservation* 238: 108183.~~

~~Schauman, Santiago A., Josep Peñuelas, Esteban G. Jobbágy, and Germán Baldi. 2024.~~ "Schleicher, Judith, Julie G. Zaehringer, Carine Fastré, Bhaskar Vira, and Terry C. H. Sunderland. 2019. "Protecting Half of the Planet Could Directly Affect over One Billion People." *Nature Sustainability* 2: 1094–96. https://doi.org/10.1038/s41893-019-0423-y.

~~The Geometry of Global Protected Lands." *Nature Sustainability* 7 (1): 82–89. https://doi.org/10.1038/s41893-023-01243-0.~~

~~Schiavina, M., S. Freire, and K. MacManus. 2015. "GHS Population Grid, Derived from GPW4, Multitemporal (1975, 1990, 2000, 2015)." Ispra, Italy: European Commission, Joint Research Centre, JRC Data Catalogue. https://ghsl.jrc.ec.europa.eu/data.php.~~

~~Skiles, Martha Priedeman, Clara R. Burgert, Siân L. Curtis, and John Spencer. 2013. "Geographically Linking Population and Facility Surveys: Methodological Considerations." *Population Health Metrics* 11 (1): 14. http://www.biomedcentral.com/content/pdf/1478-7954-11-14.pdf.~~

~~Slowikowski, Kamil. 2023. "Ggrepel: Automatically Position Non-Overlapping Text Labels with 'Ggplot2'." https://CRAN.R-project.org/package=ggrepel.~~

Stevens, Forrest R., Andrea E. Gaughan, Catherine Linard, and Andrew J. Tatem. 2015. "Disaggregating Census Data for Population Mapping Using Random Forests with Remotely-Sensed and Ancillary Data." *PloS One* 10 (2): e0107042. https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0107042.

Thomson, Dana R, Douglas R Leasure, Tomas Bird, Nikos Tzavidis, and Andrew J Tatem. 2022. "How Accurate Are WorldPop-Global-Unconstrained Gridded Population Data at the Cell-Level?: A Simulation Analysis in Urban Namibia." *Plos One* 17 (7): e0271504.

UNEP-WCMC. 2026. "Protected Area Profile for India." https://www.protectedplanet.net/country/IND.

UNEP-WCMC and IUCN. 2021. *Protected Planet Report 2021*. Cambridge, United Kingdom; Gland, Switzerland: UNEP-WCMC; IUCN.

———. 2024. *Protected Planet Report 2024*. Cambridge, United Kingdom; Gland, Switzerland: UNEP-WCMC; IUCN.

UNEP-WCMC, and IUCN. 2023. *Protected Planet: The World Database on Protected Areas (WDPA)*. UNEP-WCMC and IUCN. Cambridge, UK. www.protectedplanet.net.

United Nations Department of Economic and Social Affairs. 2021. *State of the World's Indigenous Peoples*. 5th ed. New York: United Nations. https://www.un.org/development/desa/indigenouspeoples/publications/state-of-the-worlds-indigenous-peoples.html.

Vaggi, Gianni. 2017. "The Rich and the Poor: A Note on Countries' Classification." *PSL Quarterly Review* 70 (279). https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3100742.

~~Wickham, Hadley. 2016. "Ggplot2: Elegant Graphics for Data Analysis." https://ggplot2.tidyverse.org.~~

~~Wickham, Hadley, Mara Averick, Jennifer Bryan, Winston Chang, Lucy D'Agostino McGowan, Romain François, Garrett Grolemund, et al. 2019. "Welcome to the Tidyverse" 4: 1686. https://doi.org/10.21105/joss.01686.~~

~~Wilke, Claus O. 2024. "Cowplot: Streamlined Plot Theme and Plot Annotations for 'Ggplot2'." https://CRAN.R-project.org/package=cowplot.~~

~~Wittemyer, George, Paul Elsen, William T. Bean, A. Coleman O. Burton, and Justin S. Brashares. 2008. "Accelerated Human Population Growth at Protected Area Edges." *Science* 321 (5885): 123–26. https://doi.org/10.1126/science.1158900.~~

~~WorldPop, and CIESIN. 2020. "Population Counts 20002020 UN-Adjusted Unconstrained 100m." www.worldpop.org/doi/10.5258/SOTON/WP00660.~~

1

~~Appendix~~Wani, Zubair A., Lal Singh, Sanjay Uniyal, Sunil Kumar Rana, Indra D. Bhatt, and Sunil Nautiyal. 2025. "Improving Ecosystem Vitality in India: Overcoming Barriers to Meet National and International Targets." *Environmental Sustainability* 8 (1): 17–29. https://doi.org/10.1007/s42398-025-00339-x.

West, Paige, James Igoe, and Dan Brockington. 2006. "Parks and Peoples: The Social Impact of Protected Areas." *Annual Review of Anthropology* 35 (1): 251–77. https://doi.org/10.1146/annurev.anthro.35.081705.123308.

Zhang, Tong, Luke Gibson, Jianzhang Ma, Rachakonda Sreekar, and David Lindenmayer. 2025. “Global Patterns of Border Protected Areas Reveal Gaps in Transboundary Conservation Efforts.” *One Earth*. https://doi.org/10.1016/j.oneear.2025.101206.

~~Table 2: PA Coverage and Population Proximity (2000-2020)~~

~~Percentage of land and population within and near protected areas~~

	% land within (WDPA)				% population (GHSL)				% population (WorldPop)			
	within PAs		10km of PAs		within PAs		10km of PAs		within PAs		10km of PAs	
Country	2000	2020	2000	2020	2000	2020	2000	2020	2000	2020	2000	2020
Total	**9.3**	**12.9**	**21.0**	**27.2**	**0.9**	**1.8**	**16.4**	**21.8**	**1.8**	**2.7**	**16.2**	**21.8**
Total without India	**10.2**	**14.1**	**22.9**	**29.5**	**1.4**	**2.8**	**26.5**	**33.5**	**2.8**	**4.2**	**26.1**	**33.6**
Afghanistan	0.1	3.7	0.6	6.0	0.1	1.2	0.6	14.9	0.1	1.1	0.6	13.6
Angola	5.4	10.8	9.0	15.8	0.4	1.0	2.5	4.3	0.9	1.3	3.0	5.0
Bangladesh	4.0	4.5	9.8	17.8	0.3	0.5	5.0	18.8	0.6	1.0	4.6	18.3
Benin	23.7	23.7	54.9	54.9	1.7	1.8	25.7	25.7	5.2	5.7	25.8	26.5
Bhutan	38.5	51.1	72.0	80.5	4.4	4.5	55.2	56.1	11.2	11.6	56.2	58.4
Bolivia	20.9	23.3	33.4	38.4	22.2	31.0	61.3	68.8	21.1	25.8	58.3	63.0
Burkina Faso	14.2	16.6	33.3	38.8	3.3	4.0	31.0	42.7	6.1	7.8	31.5	45.8
Burundi	5.4	5.4	40.8	41.4	0.3	0.3	40.1	44.7	2.8	2.6	39.6	43.9
Cabo Verde	0.0	18.4	0.0	94.9	0.0	8.8	0.0	96.0	0.0	14.1	0.0	94.6
Cambodia	14.1	39.9	34.3	68.5	2.1	5.3	13.5	30.7	2.4	7.7	13.6	31.3
Cameroon	5.2	11.1	13.0	25.4	0.1	3.3	4.5	25.1	0.9	6.0	5.2	24.4
Central African Republic	14.0	14.0	21.8	21.9	0.6	0.6	19.8	21.3	2.0	1.8	22.2	23.1
Chad	11.4	13.9	14.7	18.4	3.8	3.8	11.8	13.6	6.5	7.7	11.7	16.8
Comoros	0.8	33.0	6.5	97.8	0.5	11.9	36.1	99.5	1.7	38.7	21.7	98.9
Congo, Dem. Rep.	9.0	14.9	14.9	23.0	2.5	6.1	23.3	28.1	3.9	7.3	22.7	24.5
Congo, Rep.	9.3	36.5	16.3	53.1	0.7	13.2	49.2	74.3	3.3	26.9	31.7	66.4
Côte d'Ivoire	21.5	21.7	69.5	70.0	5.1	5.6	69.7	71.6	10.0	9.3	71.4	72.6
Djibouti	0.0	1.3	0.0	8.3	0.0	0.0	0.0	1.4	0.0	0.8	0.0	7.8
Egypt, Arab Rep.	4.3	11.2	6.5	15.7	0.1	0.3	12.6	13.4	0.4	0.5	11.2	11.3
El Salvador	0.1	8.6	8.2	80.7	0.0	2.9	4.2	90.2	0.0	3.8	4.2	90.3
Eritrea	0.0	0.0	0.0	1.3	0.0	0.0	0.0	0.9	0.0	0.0	0.0	1.6
Eswatini	4.1	4.4	28.3	38.4	0.2	0.1	34.1	41.8	1.6	1.5	30.7	39.5
Ethiopia	10.4	17.0	17.6	27.2	3.2	7.0	9.8	17.4	3.8	7.7	9.8	17.8
Gambia, The	5.1	8.0	32.2	61.8	0.2	1.0	57.2	79.2	2.8	8.8	55.8	78.9
Ghana	14.7	14.9	56.9	57.2	-1.2	-0.2	68.2	69.0	6.2	5.0	69.1	68.9
Guinea	7.9	23.7	35.9	52.3	2.6	14.9	46.7	57.1	4.0	15.4	47.3	59.0
Guinea-Bissau	18.0	24.4	40.8	49.9	2.8	5.8	16.7	20.7	5.8	9.4	20.9	25.3
Haiti	2.2	7.4	16.6	46.0	0.9	3.3	10.4	50.8	1.0	3.2	10.8	54.4
Honduras	20.4	23.0	54.6	63.7	4.0	5.6	67.5	76.3	8.2	9.2	65.5	74.1
India	0.2	0.5	1.1	2.4	0.0	0.0	0.5	1.1	0.1	0.1	0.5	1.1
Indonesia	11.5	11.9	31.5	33.9	0.2	0.5	32.1	34.6	1.7	2.2	31.7	34.6
Kenya	10.8	12.7	32.4	38.2	2.2	2.7	57.0	59.4	4.0	4.2	59.4	60.7
Kiribati	0.0	4.6	0.0	4.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Korea, Dem. Rep.	0.0	0.0	0.0	1.1	0.0	0.0	0.0	0.4	0.0	0.0	0.0	0.4
Kyrgyz Republic	6.4	6.7	21.6	23.3	0.6	0.6	17.0	16.7	1.2	1.1	15.1	14.7
Lao PDR	8.9	15.9	23.1	40.0	1.4	3.2	19.2	31.0	1.9	3.8	18.7	28.6
Lesotho	2.0	23.5	19.0	40.1	0.3	8.6	10.9	27.2	0.8	9.9	11.2	25.8

	% land within (WDPA)				% population (GHSL)				% population (WorldPop)			
	within PAs		10km of PAs		within PAs		10km of PAs		within PAs		10km of PAs	
Country	2000	2020	2000	2020	2000	2020	2000	2020	2000	2020	2000	2020
Liberia	1.7	3.7	6.6	12.3	0.0	0.9	3.7	10.0	0.2	1.8	3.5	10.2
Madagascar	4.3	12.8	14.0	35.8	0.2	4.8	7.7	41.7	1.6	8.0	8.4	40.2
Malawi	18.0	18.7	61.2	63.2	1.9	2.0	63.3	63.9	2.2	2.5	62.6	62.4
Mali	3.3	7.5	9.1	15.9	3.4	12.1	36.7	50.7	4.5	11.9	36.2	52.0
Mauritania	0.6	0.6	1.0	1.0	0.4	0.1	1.2	0.8	0.4	0.1	1.1	0.7
Micronesia, Fed. Sts.	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Moldova	9.1	9.1	86.1	86.4	1.4	1.6	91.5	91.5	6.3	6.0	83.2	82.1
Mongolia	14.8	19.7	23.4	32.1	1.7	2.2	36.2	54.4	5.5	6.1	33.1	55.5
Morocco	0.6	2.2	2.6	11.9	0.1	1.0	6.3	20.4	0.2	1.5	5.8	19.5
Mozambique	21.1	29.4	33.2	42.9	3.6	8.8	11.5	28.8	4.5	9.8	11.9	28.2
Myanmar	1.6	6.6	7.2	18.1	0.2	0.6	7.7	12.0	0.4	1.1	8.0	12.7
Nepal	18.9	23.5	35.2	42.8	2.6	5.5	24.4	35.8	2.9	5.6	21.2	41.6
Nicaragua	14.0	21.5	42.2	55.8	2.5	5.4	65.0	70.3	5.5	9.3	65.0	70.5
Niger	7.4	17.0	9.5	20.7	2.1	5.0	5.0	10.3	2.7	5.3	6.0	10.7
Nigeria	12.0	13.3	57.6	59.4	1.5	2.2	51.0	51.7	4.4	4.6	53.0	54.0
Pakistan	7.2	7.7	15.2	17.1	1.3	1.4	11.3	11.4	1.7	1.7	10.8	11.1
Papua New Guinea	2.9	3.7	8.6	11.5	1.2	1.2	11.1	11.8	1.5	1.7	11.6	12.7
Philippines	8.7	15.9	30.3	54.6	3.5	4.7	35.5	57.1	3.6	5.2	36.3	58.3
Rwanda	8.7	9.3	29.7	41.3	0.0	0.2	21.0	30.7	0.3	0.7	21.5	31.5
Senegal	25.5	26.5	60.1	61.4	3.1	4.1	39.5	62.1	7.6	9.4	40.8	63.4
Sierra Leone	9.4	13.0	50.3	60.1	2.5	5.6	59.2	68.5	5.8	9.1	58.9	68.7
Solomon Islands	1.0	1.5	2.1	11.6	5.2	5.7	4.9	3.6	0.1	0.2	2.4	8.1
Somalia	0.0	0.0	0.1	0.2	0.0	0.0	0.2	0.2	0.0	0.0	0.1	0.1
Sudan	1.4	1.4	1.8	2.0	0.1	0.1	0.2	0.2	0.4	0.3	0.6	0.5
Syrian Arab Republic	0.0	0.1	0.2	0.9	0.0	0.0	0.1	0.6	0.0	0.0	0.1	0.6
São Tomé and Príncipe	0.0	32.5	0.0	93.8	0.0	0.1	0.0	49.5	0.0	9.1	0.0	58.2
Tajikistan	18.7	21.8	35.1	40.9	0.6	0.7	13.3	15.7	1.5	1.7	12.5	15.2
Tanzania	36.4	39.9	65.8	69.1	4.9	6.1	60.5	66.0	8.6	9.7	61.3	67.5
Togo	12.2	12.8	62.8	66.4	3.8	4.4	49.9	65.3	5.9	6.3	49.9	66.5
Tunisia	5.1	8.0	18.1	33.5	0.1	0.3	26.1	49.7	0.8	1.5	27.6	58.5
Uganda	13.2	15.2	72.0	74.2	2.3	2.8	85.6	86.4	6.7	7.3	86.0	87.0
Ukraine	18.1	18.1	82.5	82.5	1.2	1.2	89.5	89.9	9.6	9.2	85.6	86.4
Uzbekistan	2.9	7.9	8.1	17.0	0.1	0.3	2.4	10.0	0.9	1.1	4.9	10.6
Vanuatu	3.9	3.9	37.0	37.0	0.0	-0.1	-5.8	-11.2	0.6	0.7	48.1	55.2
Viet Nam	1.9	7.6	14.9	40.1	0.5	1.1	9.1	20.4	0.9	2.0	10.4	21.9
Yemen, Rep.	1.0	1.1	1.5	2.6	1.0	2.0	3.1	10.4	1.0	2.0	3.2	10.4
Zambia	38.1	38.9	71.3	72.1	10.5	10.6	76.4	75.6	17.4	16.2	74.2	72.7
Zimbabwe	17.8	28.2	38.1	49.8	1.6	6.8	31.6	39.8	4.4	9.1	32.0	42.8
West Bank and Gaza	NA	NA	NA	NA	0.5	0.4	60.9	62.5	NA	NA	NA	NA

	% land within (WDPA)				% population (GHSL)				% population (WorldPop)			
	within PAs		10km of PAs		within PAs		10km of PAs		within PAs		10km of PAs	
Country	2000	2020	2000	2020	2000	2020	2000	2020	2000	2020	2000	2020

Source: Analysis based on WDPA, WorldPop & GHSL data (2000-2020)

a mis en forme : Bibliographie